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WONGA: The Future of Personal Finance Management – A Machine Learning-Driven Approach for Predictive Analysis and Efficient Expense Tracking

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***Abstract*— The financial literacy of Sri Lankans is relatively low, leading to difficulties in managing personal finances. This research presents a smart solution to simplify the complexities associated with money management and assist individuals in managing their finances more efficiently to achieve better financial health without requiring a comprehensive knowledge of money management from the user. The proposed system automates personal finance management with minimal user effort, reducing manual data entry by tracking cash flow by utilizing SMS messages and expense bills to extract bank transaction data and cash expenditures. Each extracted expense will automatically be categorized into the correct expense category. The system also generates a custom budget plan for each user based on spending patterns to help them stay on the budget throughout the month and avoid irrational overspending. Furthermore, the system provides a mechanism to predict future expenses associated with upcoming events based on calendar events, allowing users to devise the most efficient budget plan and avoid facing financially unprepared events in the upcoming month. All these smart solutions are bundled up in the "Wonga" mobile application to help users make better financial decisions to achieve personal financial success.**

Keywords—Personal finance, SMS tracking, expense classification, expense prediction, budget planner, image processing, machine learning, regex, NLP, ANN, CNN

I. Introduction

This research paper aims to address the challenges that hinder effective financial management for individuals, such as improper administration of incomes and expenses, lack of conventional budget planning techniques tailored to individual spending habits, unforeseen expenses, and complexities in existing financial-related applications [1]. Financial literacy is a crucial factor in financial management, and the lack of it contributes to the struggles of a majority of people in Sri Lanka, where only 35% of adults are financially literate [2]. The global personal finance software market has seen a rapid increase in demand, but current methodologies and systems are not effective for individuals with low financial literacy. Therefore, it has become increasingly important to develop a smart personal finance management mobile application that utilizes modern technologies such as machine learning, Natural Language Processing (NLP), and gamification methods to increase users' motivations and intentions to use proper personal financial management steps.

The research paper introduces a mobile application that focuses on enhancing user experience and customer satisfaction through four proposed solutions. These solutions include tracking daily expenses using both printed and handwritten bills, tracking bank transactions through SMS messages, and implementing a smart expense categorization process to classify and analyze expense data retrieved from bill images and SMS messages without having the user specifically mention the categories under which each expense falls under, generating a custom budget plan based on user spending patterns, and tracking upcoming events on the user's digital calendar to estimate expense amounts. By reducing manual work and errors, the application aims to provide an efficient and time-saving approach to managing finances with a proper budget plan, which can lead to financial success in the users' lives. These solutions aim to reduce the manual work and errors that individuals face while managing their finances and provide efficient, time-saving ways to manage finances with a proper budget plan, which will ultimately lead to financial success in their lives.

II. Literature Review

A. Smart Expense Tracker

This literature survey focuses on the proposed implementation of text extraction and processing of bank SMS and bill images using image processing and NLP. The proposed method aims to reduce the effort required for tracking finances manually. The survey conducted among the general public revealed that many people are accustomed to keeping track of their finances manually by taking down bills in a notebook or on their phone, which is prone to human errors. Several studies have been conducted to investigate the effectiveness of automated methods for tracking expenses. One such study by Du et al. [3] proposed a system that can

extract expense information from receipts and classify them into different categories. The system utilized deep learning techniques to accurately extract text information from receipts. Similarly, another study by Liu et al. [4] proposed an expense-tracking system that utilizes machine learning algorithms to extract information from receipts and bank statements. The proposed method in this literature survey utilizes both image processing and NLP techniques to extract information from bank SMS and bill images. Several studies have been conducted on the effectiveness of these techniques in various applications. For instance, a study by Xu et al. [5] proposed a method for extracting text information from images using deep learning techniques. The method achieved high accuracy in recognizing characters in images. Another study by Chen et al. [6] proposed a method for extracting information from social media posts using NLP techniques. The proposed solution in this study aims to create a well-performing, user-friendly, and helpful expense-tracking system for users to track expenses accurately and effectively without the need for manual input.

B. *Automated Expense Classifier*

Expense data classification and analysis is a major area in financial management. In this system, expenses have been proposed to manage in an automated environment to classify them correctly into correct expense categories using Naïve Bayes text classifications. Research has been done on the effectiveness of news text classification according to a study [7] and similarly despite the fact that text classification is used in many different fields, including document organization, spam filtering, document retrieval, opinion mining, and text data filtering according to a survey conducted [8], and most importantly it has not been properly used with current financial management applications related to expenses at a satisfactory level. Therefore, it is also a major concern that paved pathways to apply text classification algorithms mainly for this proposed financial management system to make the application more efficient, and accurate with better usability by helping in finance management effectively. The major algorithm which is Naïve Bayes has a high accuracy according to the study done on machine learning algorithms on text classification [9] and has good scalability and extensibility according to another conducted study [10]. Therefore, it affects this financial management system's overall performance to give correct expense classifications and insights based on those analyzed data with proper precision by providing users with a comprehensive and in-depth view of their expenses.

C. *Custom Budget Planner*

Budgeting plays a vital role in the financial planning of every individual, as it serves as a critical step towards achieving their financial objectives. Numerous developers have made attempts to create various applications for managing budgets. In the paper [11], a user-friendly mobile application is proposed for recording and analyzing transactions. Users can input daily income and expenses under default or new categories, with the app maintaining transaction history. The app also utilizes graphical representations to compare budgets against actual income and expenses and generates statistical reports to provide an overview of the user's financial situation. In the paper [12], a computerized daily expense management system that efficiently tracks day-to-day expenses has been proposed. It eliminates the need for hardcopy output by systematically maintaining records of paid and unpaid transactions. This system enables users to easily access stored data, keeping track of their expenses in an organized and effective manner. After conducting an analysis of the existing apps, it was found that none of them provide an automated solution for generating effective budget plans based on users' spending patterns. In response to this gap in the market, our system has been developed to automatically analyze users' spending behaviors and habits over the previous months using machine learning algorithms. This analysis enables the system to create a personalized budget plan that is both appropriate and effective, allowing users to better manage their finances with confidence.

D. *Future Expense Minder*

Saving money is a major concern for many people, but less consideration is given to future expenses in life [13]. Research has shown that people are biased towards optimism and suffer financially later in life due to ignorance of pessimistic scenarios [13]. Insufficient attention toward future expenses increases the risk of experiencing household economic turmoil [13]. Calendars are used to report life events and set reminders for memorable events to come in the future [14]. A survey conducted by Tungare et al. found that 93% of participants add new events to their calendars as soon as they become aware of a new upcoming event [14]. The study [15] has also indicated that online calendars are becoming one of the richest sources of contextual information that could be used to understand human behaviors. Google Calendar is the widely used digital calendar system to plan events [16]. Anthony and Wooi proposed a deep-learning model for mood prediction using calendar events. Kim et al. [15] proposed a novel model to use online calendar data for event scheduling using deep neural models. A recent pilot study conducted by Sohn et al. [17] has proposed a model to predict future healthcare expenses of patients from their chest radiographs using deep learning techniques and regression models. The work [18] presents a calendar system that can predict the likelihood of one's attendance for scheduled future events recorded using a Bayesian model. However, no research has been done utilizing calendar data for financial decision makings. This proposed application can be impactful for potential users to make the best decisions in planning their financial future without memorizing upcoming monetary events.

III. METHODOLOGY

A. *Smart Expense Tracker*

This component of the research aimed to extract expense details from two types of bills: printed bills and hand-written bills. To achieve this, different approaches were employed for each type of bill. For printed bills, which may be photos stored in a mobile phone's file storage or captured by the mobile camera, the images were first pre-processed using various techniques such as grey-scaling, sharpening, noise removal, and opening. The pre-processed images were then passed through Pytesseract, an open-source python Optical Character Recognition (OCR) library, to extract text. The extracted text was post-processed to remove any unnecessary words or characters. For hand-written bills, a custom-built model provided by the form recognizer service was utilized to extract information accurately. For digital documents, the Named Entity Recognition (NER) model was used to

identify key expense details from the extracted text. These details were then encrypted and saved in a firebase database.

To analyze bank transactions received via bank SMS, two models were trained for different classifications of bank SMS. The dataset was manually built with relevant labels to train the model. After collecting the SMS data, it was divided into two separate CSV files (SMS and spam) and combined in the data pre-processing stage in model training. Two text pre-processing methods were employed: tokenizing and added padding (Max Length = 100). Artificial Neural Networks (ANN) were chosen as the training algorithm for all classifications used in this process. The first classification performed was bank SMS binary classification, which identified whether a read SMS text was transaction-related or not. The second classification was transaction information extraction. In creating the output labels for the classification, the dataset was utilized, and commonly performed bank transactions such as "ATM Withdrawal", "Transfer Debit", "POS/ATM Transaction", "Online Transfer Debit", and "PURCHASE" were selected. Once the model identified the SMS as a transaction type, key information like the date and amount was extracted from the SMS using two patterns. For extracting the amount and transaction type, two patterns were employed, and the information was extracted accordingly. Once the expense details were extracted, they were saved in a firebase database after encryption.

TABLE I. SELECTING THE BEST ARCHITECTURE

Purpose	Tested Architecture	Best Architecture
SMS Classification	CNN-Static, LSTM DenseNet121, RNN, ANN	ANN
Bill Classification	CNN-static, DenseNet121, ANN, SVM, Decision Tree, BERT	CNN-static
Handwritten Identification	LSTM, CNN, InceptionV3, DenseNet121, NLP-NER Modeling	NLP-NER Modeling

The above Table I show the research methodology used to determine the best architecture for three different tasks: SMS classification, bill classification, and handwriting identification. Multiple architectures were tested for each task, and their performance was analyzed to determine the best architecture. The selection of the best architecture was based on the performance of each architecture for the specific task it was being used for. The methodology used in this research involved a comprehensive evaluation process that included careful analysis and thorough analysis of the performance of each architecture.

The methodology employed to track expenses proved to be successful and the extracted expense details will be used in the other components of the research as a continuation of this study.

B. *Automated Expense Classifier*

With the aid of machine learning algorithms, this component analyses expense data and automatically classifies them into the appropriate expense categories, enabling improved financial expense interpretation. Six primary personal spending categories, including food, groceries, health care, housing and bills, personal care, and transportation, will be used to examine and classify expenses. An initial survey revealed that consumers do not have a clear understanding of how to manage their spending appropriately using a category-wise methodology, which is a key contributor to financial loss. A dataset was generated that contains expenses that were gathered from many sources, including invoices, SMS, surveys, and other kinds of media, to train the machine learning model. Each expense in the dataset contributes to the accurate classification of expenses into correct expense categories. The Naïve Bayes text classification algorithm was used to create the model once all the data has been collected. The expense dataset was imported from a CSV file, together with the relevant python libraries, such as Scikit-learn, NumPy, Pandas, Matplotlib, and Seaborn, to ensure that the model operated properly. Data from various categories of personal expenses were contained in a CSV file, and based on those various exploratory data, data pre-processing was carried out to check for missing values, eliminate duplicates, and visualize the data followed by performing tokenizing, eliminating non-alphanumeric characters, and lemmatizing the words. Following this, the pre-processed text data were divided into 0.7:0.3 training and testing sets. The data were transformed into a numerical representation using Count Vectorizer. The Multinomial Naive Bayes (MNB) classification algorithm was used to train the pre-processed already split data and the model was then assessed using the accuracy percentage and also with the generated graphs on the trained and tested set. Subsequently, after manually entering the test expenses, it was evident that the model behaves as planned and produces the desired results without any accuracy limitations.

C. *Custom Budget Planner*

This component is used to forecast users' future expenses by analyzing their spending patterns to generate a customized budget that caters to the customer's needs. To achieve this task, time series datasets spanning from 2020-2023 on daily expenses were used. These daily expense records were categorized into eight different categories, namely education, medical, food, entertainment, transport, personal care, housing/bills, and other. To forecast the amounts for each expense category, a probabilistic model known as Autoregressive Integrated Moving Average (ARIMA) was employed, which effectively captures the time-varying characteristics of the daily expense data. The acquired dataset was non-stationary and since the ARIMA model requires data to be stationary differencing method was used. To achieve the best possible accuracy in forecasting, values of the hyperparameters were optimized. The dataset was split as 0.8:0.2 for training and test data and used to train the model. For each expense category, separate ARIMA models will be used to do forecasting. Accuracy matrices such as Mean Absolute Percentage Error (MAPE), Mean Error (ME), Mean Absolute Error (MAE), and Min-Max Error (minmax) were used to evaluate the model's forecasting ability. An alternative approach involving the application of the Long Short-Term Memory (LSTM) model was examined for expense forecasting, but the model failed to give acceptable results since LSTM models require more training data to capture complex patterns over an extended period. In this application where daily expenses are utilized to forecast expenses for the next 15 days, the ARIMA model was found to be a more suitable choice. This is because the ARIMA model has a proven ability to effectively capture short-term trends and seasonality in the data, making it particularly suited for short-term forecasting tasks.

D. Future Expense Minder

This component predicts expenses based on users' calendar events. Initially, conducted a survey to identify the different types of events that people mostly plan in their personal digital calendars which are likely to incur expenses. The survey identified 21 different event categories. They are virtual events (online meetings, webinars), birthday parties, weddings, trips, grocery shopping, haircut, outings for dinner or lunch, shopping outings, get-togethers, saloon appointments, visiting grandparents, exams, health check-ups, rental fee reminders, coffee outings, volunteering activities, movie nights, doctor appointments, dentist appointments, musical shows, workouts, or gym sessions. The survey also found that 83% of the people who use digital calendar platforms for personal usage are using Google Calendar as their primary calendar. The study achieved the objective through two steps: First, developed a mechanism to track the user's digital calendar and detect the calendar events that are most likely to incur an expense for the user in the upcoming month by building a classification model using NLP. Second, build regression models to predict expense amounts for the above-identified events.

The DistilBERT model was chosen to build the classification since the DistilBERT model is a faster and smaller transformer-based model, with fewer layers and fewer attention heads than Bidirectional Encoder Representations from Transformers (BERT), and is fine-tuned for specific downstream tasks, including text encoding for text classifications. A simple web application was implemented to collect a real dataset of calendar events written in the English language planned within the year 2000 - 2023 from personal Google calendars from volunteers under their permission. A total of 523 events were collected, and each event in the dataset was labeled with one of the 21 categories mentioned above using a regex-based script. The events which were not fallen into any of the above categories, which are less meaningful, were labeled into the "Other" category. The dataset was cleaned by removing HTML codes, links, emails, and other unwanted characters in event titles and descriptions and lastly by removing records with null values. the event title and description were combined into a single text. The dataset was then tokenized to get the tokenized values with the attention mask for all the words in the entire dataset. The preprocessed dataset was split as 0.9:0.1 for training and test data and used to train the model. The encoder part of the model was utilized to extract features from the text data. The decoder part was not required since it is used for language generation. The transformer pipeline was used to run the model and tokenizer together for efficient inference. Found the best model with the highest accuracy score and less validation loss by training the model for 20 epochs.

The survey gathered expense amounts associated with above mentioned 20 event types with demographic data (age, gender, province, employability status, marital status, and monthly income) from 1492 respondents. Virtual events were excluded since 98% of the respondents reported no expense for virtual events. The remaining 20 event types were observed as the set of events that are most likely to incur expenses for Sri Lankans. 20 datasets were created using survey responses and, 20 regression models were developed to predict expenses for each event category. 8 different regression models with higher performance were selected to train each dataset and found the best suitable regression model type to predict expenses for each event type. Decision Tree Regressor and Gradient Boosted Tree Regressor were found to be the best-performing models based on the R2 score, MAE, and Mean Squared Error (MSE) as shown in Fig. 6.

IV. Result and Discussion

Accuracy, a fundamental metric in evaluating classification models, continues to be the cornerstone of performance measurement. It provides an intuitive and straightforward way to assess the effectiveness of models, and as such, remains a widely used estimate. In our study on bank SMS classification, we employed two models tailored to specific use cases and split the train-test dataset into an 8:2 ratio. The accuracy of each model is presented in Table II below, reflecting the percentage of correctly identified situations out of the total samples used. These results offer valuable insights into the performance of our models and can inform future research and real-world applications in this domain.

TABLE II. RESULTS OF BANK SMS CLASSIFICATION

Model	Accuracy
Bank SMS Binary Classification Model	97%
Transaction Information Extraction Model	88%

The image preprocessing techniques utilized in our system have demonstrated exceptional results, surpassing other combinations in accuracy and output quality. Our approach has proven particularly effective in extracting text from preprocessed images, which proved superior to images without preprocessing. Given the variability in handwriting and bill structures across local stores, our two models for bill classification and extraction were tailored to address these challenges. Interestingly, our NLP-NER model outperformed regular expressions, commonly used in previous studies for key information extraction. The train-test dataset was divided into an 8:2 ratio, and the accuracy of our models for bill image information extraction is presented in Table III. These promising findings highlight the potential of our approach for future research and real-world applications in the field of automated bill classification and extraction.

TABLE III. RESULTS OF BILL EXTRACTION

Model	Accuracy
Bill Binary Classification Model	96%
Bill Information Extraction Model	82%

In expense classification, the accuracy of the model is a must for the system to perform well. According to the trained model using the text classification algorithm used, the results of the analysis showed that the MNB classifier trained on the pre-processed text data has been able to achieve an accuracy of 98.93% on the testing set.

The classification report which provided additional insight into the classifier's effectiveness demonstrated how precision, recall, and F1-score differed comparably amongst various expense categories. Overall, the final results and outcomes demonstrated that there is still potential for enhancement, notably for differentiating between specific spending categories, even though the pre-processing

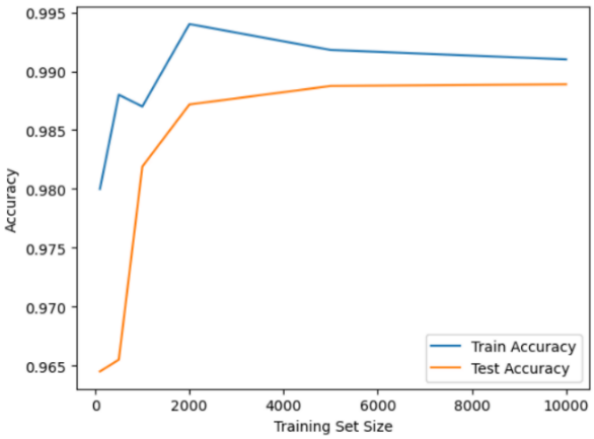

Fig. 1. Accuracy Graph on Text Classification

processes optimized the classifier's performance compared to using raw text data.

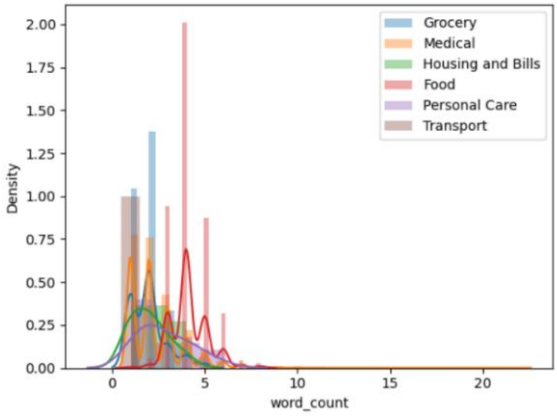

Fig. 2. Density vs Word Count Graphs for Categories

According to the model outcome analysis, it was evident and emphasized the significance of precise choosing and pre-processing text data when developing a text classification model. It also emphasized the possible advantages of using more sophisticated models or feature engineering techniques to further enhance performance.

When considering how to forecast expenses for creating a tailored budget plan, two different models, ARIMA and LSTM, were assessed.

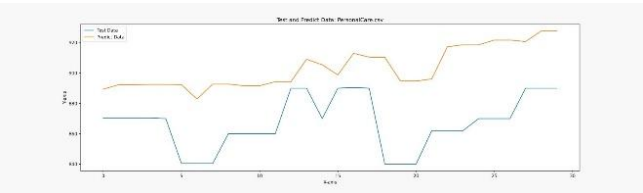

Fig. 3. Test and Predict Data Graph - LSTM Model

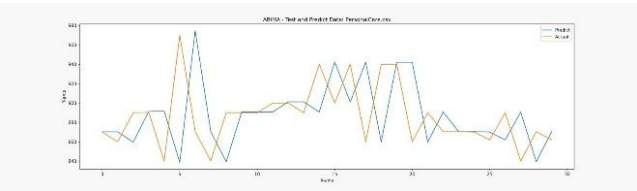

Fig. 4. Test and Predict Data Graph - ARIMA Model

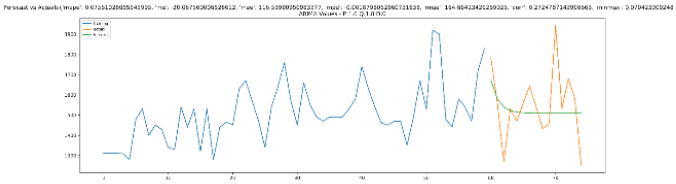

Fig. 5. Forecast vs Actual Value Graph ARIMA Model

After making the datasets stationary, the ARIMA model outperformed the LSTM model in forecasting future values for expense categories. Specifically, the ARIMA model was able to forecast values for the personal care category with a MAPE of 0.0756, which indicates that, on average, the model's predictions have an error rate of approximately 7.56%, an acceptable rate in this context. The reason for ARIMA's superior performance is that it works well with small datasets, whereas LSTM models require large datasets to learn complex patterns in the data. Since the collected time series data for expense categories were limited, the ARIMA model was better at calculating the next predictions. Therefore, the stationary ARIMA model proved to be a more suitable choice for forecasting future expense values for the budget plan.

Regarding the future expense minder component of the study, the DistilBERT model achieved 92% accuracy and 0.68 validation loss after being trained for 20 epochs. This proves that the model is effective in classifying event categories for calendar events, enabling the system to recognize the events that are most likely to incur expenses for the user.

Regression analysis accurately predicted future expenses for most event categories, with accuracies ranging from 80% to 90%, except for exams since it is less likely to incur expenses for exams in general. The R2 scores obtained for each regression model are shown in Fig. 6. The higher R2 scores for each expense predictor model showed that the models effectively predicted future expenses for upcoming events based on user demographic data.

Event Category	Max. Score given Reg. Type	R2 Score	Mean Ab. Error	Sq. Mean Ab. Error
Birthday party	DecisionTreeRegressor	0.86	443.08	1550.87
Trip	DecisionTreeRegressor	0.87	505.21	1538.26
Shopping outing	DecisionTreeRegressor	0.84	949.82	2679.22
Health check-up	DecisionTreeRegressor	0.85	585.69	1915.68
Coffee outing	DecisionTreeRegressor	0.8	175.11	419.47
boarding/hostel fees	DecisionTreeRegressor	0.85	1010.38	2082.12
grocery shopping	DecisionTreeRegressor	0.86	883.76	3107.61
visiting grand parents	DecisionTreeRegressor	0.98	280.86	799.62
haircut	GradientBoostingRegressor	0.91	309.35	519.68
get together	DecisionTreeRegressor	0.89	311.08	783.95
Exam	DecisionTreeRegressor	0.16	2220.68	13877.98
Saloon appointment	DecisionTreeRegressor	0.92	248.31	613.46
charity	DecisionTreeRegressor	0.98	169.48	473.88
movie night	DecisionTreeRegressor	0.91	198.93	541.55
doctor appointment	DecisionTreeRegressor	0.9	239.15	695.23
dentist appointment	DecisionTreeRegressor	0.88	201.51	612.83
musical show	DecisionTreeRegressor	0.97	266.36	597.01
Wedding	GradientBoostingRegressor	0.7	996.29	1766.97
lunch/dinner out	DecisionTreeRegressor	0.95	287.46	791.09
workout/gym	DecisionTreeRegressor	0.92	320.5	669.51

Fig. 6. Performance of Event Expense Predictor Models

V. Conclusion

In-depth analysis by Wonga financial application research led to the recommendation of an intelligent solution in the form of a mobile application that automates personal finance management with minimal user effort, simplifies money management complexities, and helps people manage their finances more effectively. This solution is specifically designed for Sri Lankans who frequently struggle with effectively managing their finances because of their inadequate level of financial literacy. The application uses sophisticated technologies like image processing, machine learning, and natural language processing as conventional support for that to improve performance with better accuracy and with novelty in features of the system that encourage users to adopt proper personal financial management steps. The proposed solutions concluded the developments to reduce manual work and errors using automating manual processes by giving solutions like text extraction from bank SMS, bill images, automatedly classifying expenses, providing efficient and time-saving ways to manage finances, giving predictions on future expenses based on calendar events, and generate a custom budget plan based on user spending patterns. Since currently available applications do not provide adequate assistance that covers all aspects of financial management in a single application or a system, individuals may be able to improve their financial health and personal financial success by using this sophisticated approach rather than using multiple systems or applications for each task. By thoroughly examining a variety of factors, this study makes a significant contribution to improving personal financial literacy and efficient financial management in Sri Lanka and beyond.

The current work proposes an expense tracking and prediction application that could be scaled to the organizational level by integrating it with existing financial systems. The app is limited to English language documents, and future research could improve its accuracy for other languages. Additionally, future research could explore the use of location data to predict travel costs more accurately, considering factors such as local currency exchange rates and transportation costs. These potential research directions could enhance the usefulness of the proposed application for both individuals and organizations.

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